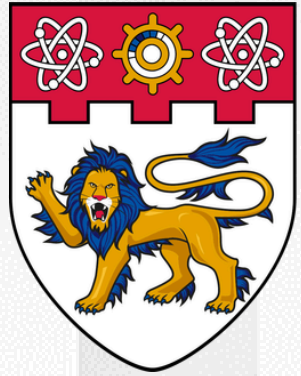




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FairCredit Jurisdiction Navigator

A jurisdiction-aware AI credit decisioning compliance tool for American Express

Presenter: LI CHENGUANG



Why this Matters

AI credit decisions create different compliance questions across jurisdictions

BUSINESS PROBLEM

American Express can use AI-supported credit decisioning to improve speed and risk segmentation. But a model score alone does not answer the compliance questions that matter to regulators, consumers or senior management.

US QUESTION

Can we explain this declined applicant?

EU QUESTION

Is the AI system governed and monitored?

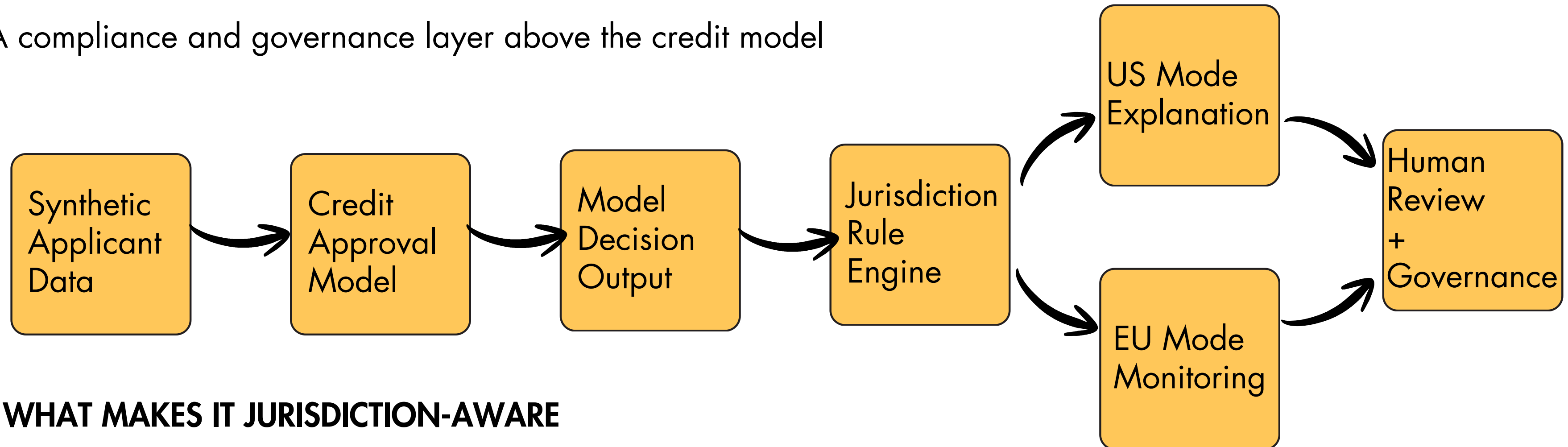
TOOL ANSWER

Interpret model outputs by jurisdiction

Dimension	United States	European Union
Regulatory focus	Adverse action explainability	High-risk AI governance
Main output	Specific decline reasons	Fairness + drift monitoring
Human review	Declined / borderline cases	Fairness, drift and governance alerts

Tool architecture

A compliance and governance layer above the credit model



WHAT MAKES IT JURISDICTION-AWARE

1. The model produces a credit decision, but the compliance layer decides what that decision means by jurisdiction.
2. US mode emphasises specific adverse action reasons for individual applicants.
3. EU mode adds portfolio-level fairness metrics, PSI drift monitoring and governance review triggers.
4. Human review is retained for borderline cases, fairness signals and jurisdictional uncertainty.

Prototype evidence

Synthetic data + transparent model + operational decisions

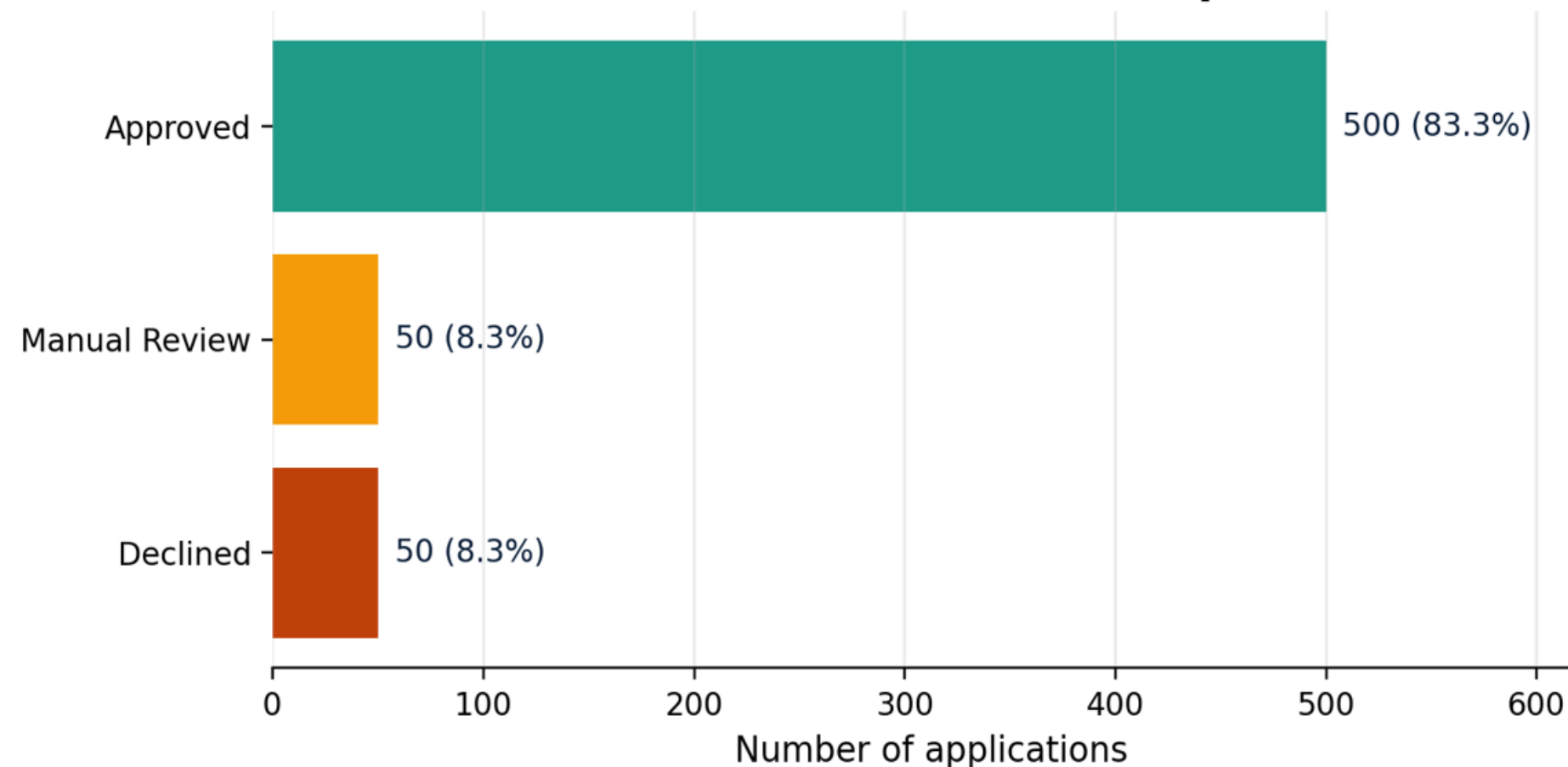
SYNTHETIC RECORDS
2000

TEST APPLICATIONS
600

MODEL
Logistic regression

GROUP PROXY
Monitoring only

Model Decision Distribution (Test Sample, n=600)



INTERPRETATION

Approved: 500 (83.3%)

Manual Review: 50 (8.3%)

Declined: 50 (8.3%)

The high approval rate reflects synthetic data and simplified thresholds — not American Express's real policy.

The purpose is to demonstrate compliance mechanics, not a production-ready credit model.

US mode: individual adverse action explanation

The key question is whether a declined decision can be explained clearly and specifically

US COMPLIANCE LOGIC

In US mode, the tool focuses on applicant-level explanation. A generic reason such as “low model score” is not sufficient for a negative credit decision.

Declined and borderline applications are routed for explanation review.

The explanation layer uses transparent, auditable rules.

Human reviewers check whether reasons are clear and specific.

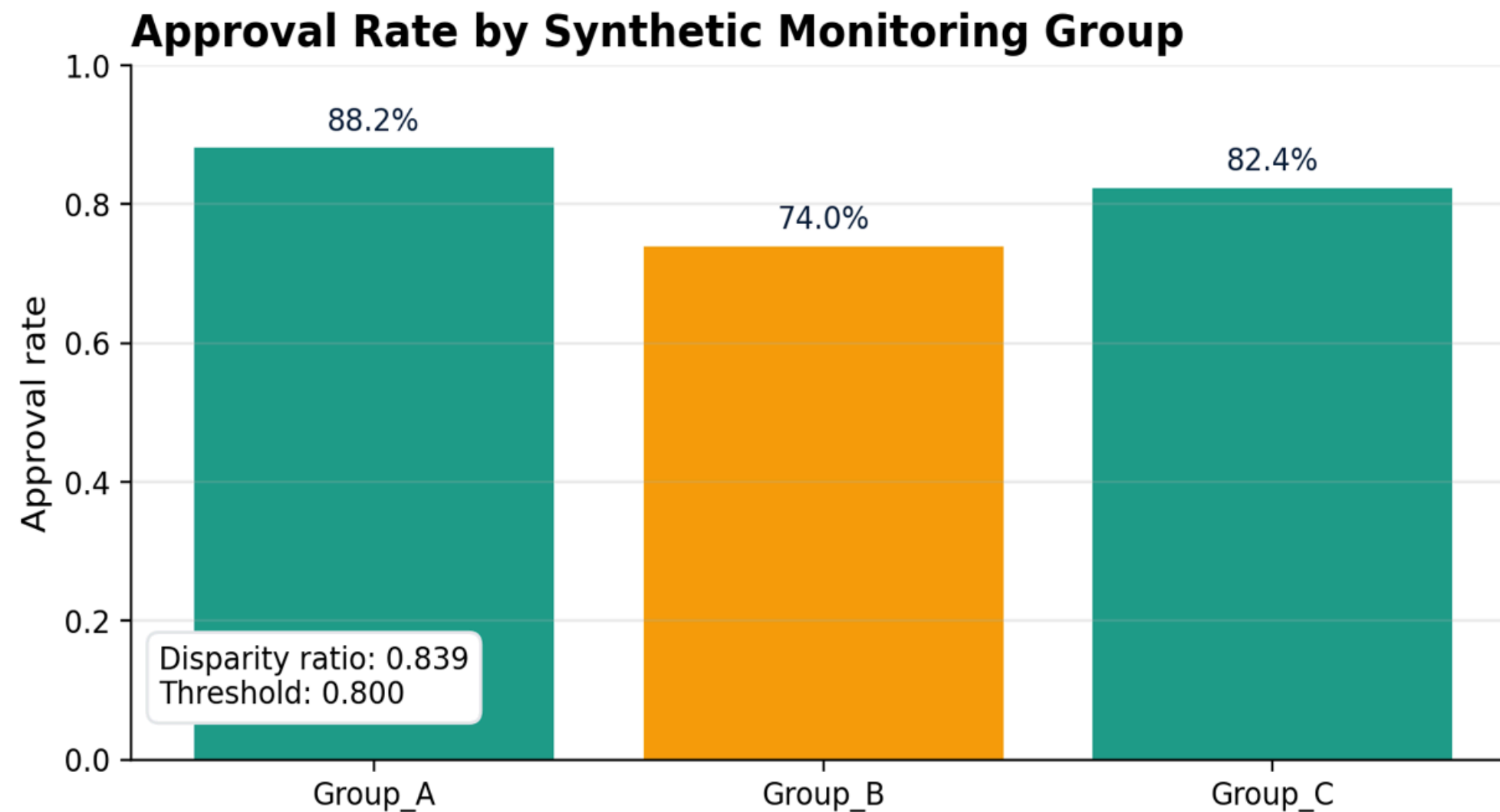
Management value: a clearer audit trail for individual negative credit decisions.

Example US Mode Output

Field	Output
Decision	Declined / Manual Review
Reason 1	High debt-to-income ratio
Reason 2	Limited credit history
Reason 3	Recent missed payments
Review	Check clarity and accuracy

EU mode: fairness monitoring

The system is monitored at portfolio level, not only at individual-decision level



LOWEST
APPROVAL
Group B
74.0%

HIGHEST
APPROVAL
Group A
88.2%

DISPARITY RATIO
0.839

ALERT THRESHOLD
0.800

INTERPRETATION

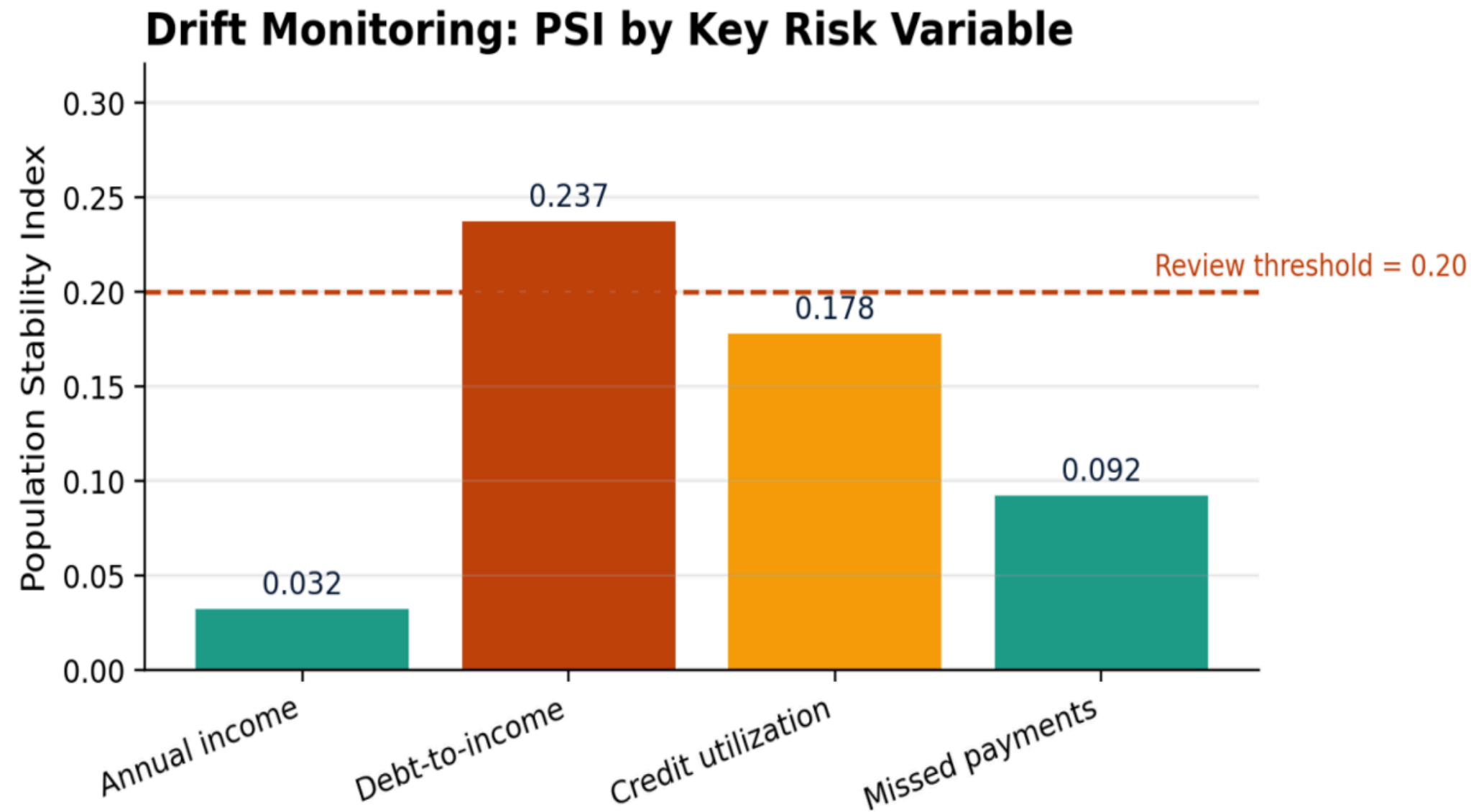
No hard fairness alert is triggered because 0.839 is above the 0.80 threshold.

Group B still shows a noticeably lower approval rate and average model probability.

This is treated as a monitoring signal rather than a clean pass.

EU mode: drift monitoring and governance review

The strongest signal is not a hard fairness breach, but a material population shift



MATERIAL DRIFT
Debt-to-income PSI = 0.237 > 0.20

MODERATE SIGNAL
Credit utilisation PSI = 0.178

GOVERNANCE IMPLICATION

1. EU mode triggers human model governance review due to material drift in a key credit-risk variable.
2. This shows why governance monitoring should not wait until a bias threshold is breached.
3. The tool helps senior management see when model assumptions may no longer fit the current applicant population.

Senior management takeaways

What this prototype proves — and what it does not claim

WHAT THE TOOL PROVES

1. A jurisdiction-aware layer can convert one model output into different compliance workflows.
2. US mode produces specific adverse action reasons for individual negative decisions.
3. EU mode adds portfolio-level fairness, drift and human governance review.
4. Drift monitoring triggered review even without a hard fairness breach.

CLEAR BOUNDARIES

1. Uses synthetic data only; no real American Express data.
2. Does not reproduce the real American Express credit model or policy.
3. Does not provide legal advice or final approval decisions.
4. Thresholds are illustrative and require validation before deployment.

Recommended next steps

Add SHAP-based explanations and reviewer workflow

Introduce versioned jurisdiction rules with audit logs

Validate thresholds with legal, compliance and model risk teams

Expand to UK Consumer Duty or Singapore MAS AI governance in a later version

References and data note

Key sources used to frame the regulatory logic

Consumer Financial Protection Bureau. (2023). CFPB issues guidance on credit denials by lenders using artificial intelligence.

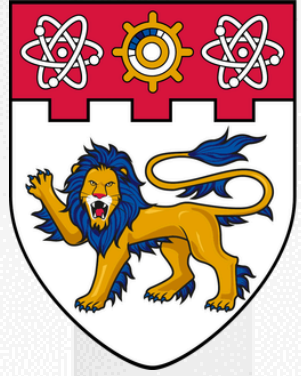
European Union. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence.

National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0).

Office of the Comptroller of the Currency. (n.d.). Financial Institution Search: American Express National Bank. Task 3 prototype outputs: synthetic data, scored applications, fairness metrics and PSI drift monitoring.

Data note

All quantitative results shown in this deck are generated from synthetic credit application data for prototype demonstration. No real American Express customer data is used, and the model is not production-validated.



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Thanks for listening!

